

ROHAN W

Bengaluru, India

+91 9840571093

rohan.w.charles@gmail.com

linkedin.com

github.com

Technical Skills

- **LLM & GenAI:** Fine-tuning (LoRA/QLoRA/PEFT), Prompt Engineering (DSPy, CoT, Meta-Prompting), Multi-agent Systems, RAG
- **Serving & Optimization:** vLLM, CUDA, CUDA Graphs, Dynamic Batching, Model Quantization, TensorRT, ONNX, LLM Inference Optimization
- **ML Frameworks:** PyTorch, TensorFlow, LangChain, Scikit-learn, Pandas, NumPy
- **Infrastructure & MLOps:** Temporal, Celery, Docker, Kubernetes, Helm, CI/CD, A/B Testing, Multi-cloud Architecture

Experience

Senior Machine Learning Engineer

March 2025 – Present

Avalara

Bangalore, IN (Remote)

- Engineered and deployed an internal, multi-cloud LLM server providing a secure, standardized **OpenAI-compatible API** across **Azure, AWS Bedrock, and Google Vertex AI** for all agent development teams company-wide. Handles **1 million requests daily**, reducing engineering effort through a unified abstraction layer while ensuring model flexibility and strict data governance.
- Built and deployed a custom GPU inference pipeline for a **BERT-based HS code multi-label classification** system, implementing **custom CUDA optimizations**, dynamic batching, and efficient scheduling to improve **latency, throughput, and GPU utilization** in production.
- Improved ML operational reliability and scalability by leading the infrastructure migration of long-running evaluation and prompt finetuning jobs from **Celery to Temporal**, enhancing workflow management and fault tolerance.

ML Research Engineer - II (L5)

April 2024 – March 2025

Amazon

Bangalore, IN

- Improved Items Glance View by **20% across 100 million products** through **A/B testing** for products without defects. Built POC using Claude, then deployed production system using **fine-tuned Mistral 7B model with Ray and vLLM** to increase accuracy.
- Developed platform using LLMs (Claude 3.5 Haiku) to accelerate Data Associates' productivity for adding missing attributes to the Amazon catalog, significantly boosting data backfilling efficiency.
- Developed and presented Platform Proof-of-Concept for Automated Prompt Finetuning using DSPy to leadership, projected to reduce manual prompt engineering effort and boost ML Engineer productivity by 140%, helping the team reach Q4 goal faster. Recognized with **RISE Q4 2024 award**.
- Enhanced Amazon retail search experience utilizing advanced prompting techniques including Chain-of-Thought (CoT) and Meta-Prompting across internal services.

Senior Machine Learning Engineer

Feb 2020 – June 2024

Rapid Acceleration Partners

Chennai, IN

- Led, **mentored and guided a team of 4 members** in the Machine Learning Team
- Re-architected parts of the **RAPFlow** ML pipeline to support more efficient GPU deployment, introducing **CUDA-backed inference optimizations**, export/runtime improvements, and batch processing changes to improve production efficiency and usability.
- Upgraded the in-house OCR inference stack using **CUDA/TensorRT-based optimization**, reducing inference time by **48%** without compromising output quality.
- Developed and implemented a chatbot powered by GPT-3.5 and LangChain to automate customer interactions and lead them to setup an appointment with the car dealership. The chatbot was able to answer customer questions, provide product information, and schedule appointments. As a result, the chatbot increased customer satisfaction, reduced drop off, and improved the sales funnel
- Lead a team that built and deployed AI driven automation tools to optimize sales leads for US car dealerships as well as create new sales leads to drive revenue growth. In addition to that we developed a tool to optimize warehouse workflow
- Built GPU-optimized document classification and document understanding pipelines for the airplane-parts domain, using **CUDA/TensorRT/ONNX-based inference optimization** and batching strategies to improve **latency and throughput** in production.

Data Science Associate

Dec 2018 – Nov 2019

Zoomrx

Chennai, IN

- Developed data processing tool so as to ensure that data from different sources integrates seamlessly with the different modules that consume that data

Education

SRM University

July. 2015 – July 2019

B.Tech E.C.E

Chennai, TN

OSS Contributions

- Active contributor to widely adopted open-source ML and systems libraries, with merged pull requests across core Python runtime, LLM infrastructure tooling, and research-driven optimization frameworks; representative selection shown below, with full contribution history available on GitHub.

Certifications

Deep Learning Specialization

Nov 2019

Coursera

Game Theory

Sept 2020

Stanford University